Course:	Object Oriented Programming	Course Code:	CS-217
Program:	BS(Computer Science)	Semester:	Spring 2020
Duration:	60 Minutes	Total Marks:	25
Paper Date:	26-Feb-2020	Pages:	4
Section:	ALL	Section:	
Exam:	Midterm Exam 1	Roll No:	

Instruction/Notes:

1. Answer in the space provided
2. You can ask for rough sheets but **they will not be graded or marked**
3. In case of confusion or ambiguity make a reasonable assumption.
4. Questions are not allowed.

Question 1: (5+5 marks)

Write output against each of the following in proper format? (Write **G** for garbage value, if any).

Part(A) (5 marks)

```
void Interchange(int*& a, int*& b)
{
        int *temp = a;
        a = b;
        b = temp;
}
```

```
void print(int **a, int size) {
        for (int **i = a; i < a + size; i++){
                for (int j = 0; j < size; j++)
                        cout << *(i[0] + j) << ",";
                cout<<endl;
        }
}
```

```
int main() {
        int size = 3;
        int ** a = new int *[size];
        for (int i = 0; i < size; i++)
        {
                *a = new int[size];
                for (int j = 0; j < size; j++){
                        *(*a + j) =i+ j + 1;
                }
                int index = (i + 1) % size;
                Interchange(*a, a[index]);
        }
        print(a, size);

        for (int i = 0; i < size; i++)
                delete[] a[i];
        delete[] a;
        a =nullptr;

        return 0;
}
```

OUTPUT:

Part(B) (5 marks)

```
class A {
        char c1, c2;
public:
        A()
      {     c1 = 'z';
            cout << c2 << ",";
        }

        A(char c)
      {         c2 = 'A';
                cout << c << ",";
        }
        void setC1(char n1) { c1 = n1; }
        void setC2(char n2) { c2 = n2; }
        void swap() { c1 = c2; c2 = c1; }
        char getC1() { return c1; }
        char getC2() { return c2; }
};
```

```
void main() {
        A a1;
        A a2('F');

        cout << endl;
        cout << a1.getC1() << "," << a2.getC2() << endl;

        a1.setC1('O');
        a2.swap();

        cout << a1.getC1() << "," << a1.getC2() << endl;
        cout << a2.getC1() << "," << a2.getC2() << endl;

        a1.swap();
        cout << a1.getC1() << "," << a1.getC2() << endl;
}
```

OUTPUT:

Question 2: (3+3+9 marks)

You have to implement the C++ code of a function **splitArray,** that takes a dynamic square two-dimensional array ***Arr*** and its size **n** as parameters. Array ***Arr*** comprises of random integer numbers. The function **splitArray** will split array ***Arr*** in two sub arrays and return them along their sizes. The first sub array will contain all prime numbers of original ***Arr*** and second sub array will contain all non-prime numbers.

Sample run of **splitArray** is shown below:

Input Arr:	Sub Arrays returned by splitArray:
Enter a Number greater than 1: 5 Original Array: 38 19 38 37 55 97 65 85 50 12 53 0 42 81 37 21 45 85 97 80 76 91 55 6 57	Sub Array of Primes: 19 37 97 53 37 97 Sub Array of Non-Primes: 38 38 55 65 85 50 12 0 42 81 21 45 85 80 76 91 55 6 57

Note:

1. The Original array **Arr** should remain intact (there should be no change in its size and data) after function call.
2. The function **bool isPrime (int n)**, which returns true if a number is prime and false otherwise is already implemented. **So you do not need to implement it**.
3. Your code should be free of dangling pointers and memory leak.

Part (A) Write down the function header of **splitArray**. **(3 marks)**

Part (B) Write the C++ code of a generic function **deallocateArray**, which can deallocate memory of 2d arrays of any size. This function will be called from main function **three** times for deallocation of original array ***Arr***, and two sub arrays which are created and returned by function **splitArray**. **(3 marks)**

Part (C) Write the C++ code of function **splitArray**. **(9 marks)**